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Evaluating Agricultural Loans and Expenditure Allocation in Nigeria Using Double Machine Learning

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Abstract

Agricultural credit is widely promoted as a policy instrument for advancing food security and agricultural transformation in low- and middle-income countries. Yet evidence on how smallholders allocate credit they receive remains mixed, particularly in contexts characterized by limited formal financial penetration. This paper examines how agricultural credit affects household expenditure allocation in Nigeria using panel data and a Double/Debiased Machine Learning (DML) framework. We estimate the causal impact of receiving an agricultural loan on farm and nonfarm expenditures and allow treatment effects to vary continuously with farm size. On average, credit increases farm and nonfarm expenditures by 14.5 percent. However, substantial heterogeneity emerges: the farm investment response rises with landholdings, while the effect on nonfarm expenditures does not vary systematically with scale. Evidence from disaggregated outcomes suggests fertilizer intensification as one mechanism among larger smallholders. These findings indicate that agricultural credit operates through differentiated channels depending on production capacity, supporting intensification where scale permits and primarily relaxing liquidity constraints among smaller farms. The results highlight the importance of accounting for heterogeneity in the design and evaluation of agricultural finance policies.

Key Words: agricultural credit, household expenditures, Nigeria, double/debiased machine learning

JEL Classification: C45, O12, O23, Q12

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1 Introduction

Across low- and middle-income countries, agricultural credit has become a central policy instrument for advancing food security, rural development, and agricultural transformation. Governments and multilateral institutions increasingly rely on subsidized lending, credit guarantees, and risk-sharing facilities to relax liquidity constraints and promote farm investment (FAO, 2019). In principle, expanded access to agricultural finance enables farmers to invest in productivity-enhancing inputs, smooth seasonal cash flows, and manage risk.

In Nigeria, where smallholders dominate the farming population (FAO, 2018) and credit markets remain largely informal, agricultural credit occupies a prominent place in rural development policy. Programs such as the Agricultural Credit Guarantee Scheme Fund (ACGSF), the Commercial Agriculture Credit Scheme (CACS), the Nigeria Incentive-Based Risk Sharing System for Agricultural Lending (NIRSAL), and the Anchor Borrowers' Programme (ABP) aim to inject liquidity into the agricultural sector. While some initiatives have been associated with increases in aggregate production (Umeh and Adejo, 2019), credit penetration remains low: only about 5 percent of farming households report receiving loans for agricultural purposes (World Bank Group, 2025), and most loans originate from informal or semi-formal sources.

This environment raises a persistent empirical question in smallholder agricultural finance: when farmers gain access to credit, does it translate into productive farm investment, or does it primarily support consumption smoothing and risk management? Understanding how credit is deployed once received (and whether responses vary across farmers with different production endowments) remains central to evaluating credit-based interventions.

We address this question using panel data from Nigeria and a Double/Debiased Machine Learning (DML) framework. We estimate the causal impact of receiving an agricultural loan on household expenditure allocation, distinguishing between farm expenditures (inputs, land, labor, and livestock) and nonfarm expenditures (food and other household consumption). We further examine heterogeneous treatment effects as a continuous function of farm size to test whether credit responses vary systematically across the land distribution.

A large empirical literature examines whether relaxing credit constraints translates into higher agricultural investment and productivity. Early work emphasizes that credit access affects both the level and composition of farm investment under incomplete markets

(Carter and Olinto, 2003). More recent experimental and quasi-experimental studies find that credit can increase agricultural investment, but that impacts are highly heterogeneous and depend on farm scale, timing, and complementary institutions such as insurance or savings mechanisms (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2018). In many settings, larger or less constrained farms are better positioned to convert liquidity into productivity-enhancing investments, while smaller farms face structural constraints that limit returns to credit-financed input use. Consistent with this, a parallel literature shows that agricultural and microenterprise credit frequently functions as general household liquidity rather than narrowly targeted investment finance. Borrowers often reallocate funds toward consumption smoothing, risk management, or nonfarm activities, particularly in environments characterized by income volatility, food insecurity, and missing insurance markets (Angelucci et al., 2015; Banerjee et al., 2015; Meager , 2019). These patterns align with household portfolio optimization under binding constraints, where stabilizing consumption and food security may dominate marginal on-farm investment, especially among smaller or more vulnerable farms.

Our results indicate that receiving an agricultural loan increases both farm and nonfarm expenditures by 14.5 percent on average. However, these average effects mask substantial heterogeneity. The farm investment response rises with landholdings: each additional hectare increases the credit-induced farm expenditure effect by 8.2 percent. In contrast, the effect on nonfarm expenditures does not vary significantly with farm size. We further find evidence that fertilizer intensification drives this scale-dependent divergence. This is consistent with previous empirical evidence suggesting that the smallest or liquidity constrained landholders often allocate a disproportionate share of additional liquidity toward nonfarm or consumption expenditures rather than farm investment (Hussain and Thapa, 2016; Fink et al., 2018).

These findings suggest that agricultural credit operates through different channels depending on farm scale. For smaller landholders, credit primarily relaxes household liquidity constraints, while for relatively larger smallholders it supports productive intensification. From a policy perspective, the results underscore the limits of uniform credit expansions and highlight the importance of scale-sensitive credit design and complementary risk-mitigation mechanisms.

Our paper contributes to the literature on agricultural credit and smallholder behavior in three ways. First, we provide new evidence on the allocation of agricultural credit across uses in Nigeria by documenting how loan access affects both farm and nonfarm expenditures. Rather than focusing solely on production outcomes, our expenditure-based approach captures multiple channels through which credit relaxes household constraints, including consumption smoothing, risk management, and farm investment. Second, we contribute to the literature on heterogeneous treatment effects by estimating how expenditure responses vary continuously with farm size, rather than grouping farms into broad size categories. We identify a systematic relationship in which smaller landholders allocate a larger share of liquidity toward nonfarm expenditures, while farmers with greater land endowments increasingly channel credit toward productive farm investment, helping reconcile mixed evidence on the investment impacts of agricultural credit. Third, we demonstrate the applicability of Double/Debiased Machine Learning (DML) in a panel setting to recover policy-relevant, plausibly causal estimates in environments characterized by high-dimensional covariates and non-random program participation. In contexts such as Nigeria, where experimental variation is limited and credit programs are implemented at scale, our approach offers a practical framework for evaluating food system interventions while accounting for household heterogeneity.

The remainder of the paper is organized as follows. Section 2 describes the institutional context and data. Section 3 outlines the empirical strategy. Section 4 presents results on average and heterogeneous effects of agricultural credit on expenditure allocation. Section 5 discusses implications for credit design and food system governance, and Section 6 concludes.

2 Study Setting

2.1 Institutional Context

Although Nigeria has long been characterized as Africa's largest oil exporter, oil revenues have declined as a share of GDP over the past three decades. Over the same period, agriculture has become increasingly central to the domestic economy, surpassing oil in its contribution to GDP and accounting for roughly one-quarter of total employment. Despite this economic importance, the agricultural sector remains highly fragmented and dominated by smallholders, with an average farm size of approximately 0.5 hectares. Development challenges are substantial. Survey evidence suggests that nearly 70 percent of smallholder farmers live below the poverty line, most operate at subsistence levels where a large share of output is consumed rather than marketed, and only about 7 percent report access to formal credit (FAO, 2018). In theory, relaxing liquidity constraints in such settings should generate high marginal returns to agricultural investment (Ali et al., 2014; Carter and Olinto, 2003). In practice, however, binding consumption needs and income volatility may lead households to allocate additional liquidity toward food, essential goods, or risk management rather than exclusively toward farm intensification (Karlan et al., 2014; Suri et al., 2021; Bocher et al., 2017). Low credit penetration therefore presents both a constraint and a policy opportunity, and credit expansion has the potential to improve household welfare while strengthening domestic food production.

Nigeria's farm credit market is segmented across formal, public, and informal institutions. Formal providers include commercial banks, microfinance institutions, and public entities such as the Nigeria Agricultural Cooperative and Rural Development Bank (NACRDB). Informal and semi-formal mechanisms—including family networks, cooperatives, and rotating savings associations (e.g., Ajo or Esusu)—play a critical role in filling gaps left by the formal sector (Adetiloye , 2012). Nonetheless, overall credit access remains limited. Formal lenders face high transaction costs, collateral constraints, and information asymmetries that restrict lending to smallholders, while informal lenders lack sufficient capital to meet broader demand (Awotide et al., 2015; Osabohien et al., 2020; Ozoko et al., 2022; Ukwuaba et al., 2020). This segmentation reflects broader patterns documented in the credit literature: marginal enterprises operating at very small scales often prioritize immediate consumption and liquidity needs when credit becomes available, whereas larger producers are better positioned to translate borrowing into productive investment (Banerjee et al., 2015; Hussain and Thapa, 2016).

In response to these constraints, the Federal Government and the Central Bank of Nigeria have repeatedly implemented directed lending and risk-sharing programs. The Agricultural Credit Guarantee Scheme Fund (ACGSF), established in 1977, guarantees up to 75 percent of loan defaults for participating lenders. Subsequent reforms introduced mechanisms such as the Self-Help Group Linkage Banking model, the Trust Fund Model, and the Interest Drawback Scheme, all intended to reduce effective borrowing costs and encourage lending to smallholders. Other initiatives, including the Family Economic Advancement Programme, sought to expand and partially formalize traditional savings mechanisms (FAO, 2013; Obinne , 1999). Despite these efforts, credit to agriculture remained limited, averaging roughly 3 percent of total national credit between 1978 and 2006 (Adetiloye , 2012).

More recently, the Anchor Borrowers' Programme (ABP), launched in 2015, aimed to address coordination failures through an interlinked transaction model connecting banks, farmers, and off-takers. The program provided credit guarantees while requiring farmers to sell output to designated buyers to reduce default risk. At its peak, the ABP facilitated over one trillion Nigerian naira (approximately USD 2.7 billion, c. 2020) in loan commitments and has been associated with increases in domestic grain production (Umeh and Adejo, 2019). However, repayment rates were reported to be approximately 25 percent (IMF, 2023), and amid tightening fiscal conditions, new loan applications were suspended in 2024.

These developments occurred alongside substantial macroeconomic shocks. Border closures in 2019 contributed to rising food prices, followed by oil price volatility in 2020 that constrained government revenues. Subsequent fiscal adjustments included the unwinding of programs such as the ABP, the removal of fuel subsidies, and exchange rate reforms. Together, these shocks have intensified food insecurity[1] and heightened pressure on rural households. In this environment, understanding how credit is allocated once accessed is particularly salient. Distinguishing whether loans support agricultural intensification or primarily facilitate consumption smoothing is central to evaluating the effectiveness of credit-based food system interventions.

2.2 Living Standards Measurement Study Core Data

We use data from the Nigeria General Household Survey Panel (GHS-Panel), part of the World Bank's Living Standards Measurement Study–Integrated Surveys on Agriculture (LSMS-ISA). The GHS-Panel is nationally representative and designed for policy analysis. Our analysis draws on the two most recent waves (Wave 4: 2018–2019; Wave 5: 2023–2024), each covering approximately 5,000 households, with roughly 89 percent successfully re-interviewed across waves. Each wave includes household, community, and agricultural questionnaires collected during post-planting (July–September) and post-harvest (January–March) phases (NBS, 2019, 2024).

The household module records demographic characteristics, asset ownership, income sources, economic shocks, and food consumption. We aggregate individual-level information to the household level, using head-of-household characteristics as demographic controls. The agricultural module is recorded at the plot level and includes land tenure, plot size, crop production, and fertilizer use. Plot-level variables are aggregated to the household level for key measures such as total cultivated area and fertilizer use.

We construct two primary outcome variables: farm expenditures and nonfarm expenditures. Farm expenditures include land (rent or purchase), livestock (purchase and upkeep), crop inputs (seeds, fertilizer, machinery, pesticides), and hired labor. Nonfarm

[1] For the interested reader, we present evidence of the rising food insecurity among Nigerian households between the period 2018-2024 in figure A1 of the supplementary Appendix A.

expenditures include imputed food consumption and non-food expenditures (durables, non-durables, and services), excluding nonfarm business investments, education, health, vehicles, ceremonies, and holidays to isolate regular household consumption. All expenditure variables are annualized to ensure comparability across recall periods.[2]

Key control variables include farm land area, value of crop production, durable assets, livestock holdings, nonfarm income, land rights, economic shock indicators, food security measures, and demographic characteristics.

To construct the analytical sample, we harmonize variable definitions across waves and standardize categorical variables where classifications differ. Observations that cannot be matched to the agricultural module or linked across waves are excluded, reducing the sample from 9,229 to 6,398 observations (69.3 percent retention), corresponding to 3,199 agricultural households observed in both waves. For specifications involving farm-size interactions, we restrict to 5,900 observations with non-missing land area. As discussed in Section 3, nuisance functions are trained using the full paired panel with imputation to maximize predictive performance, while final treatment effect estimates are reported on the restricted subsample to maintain consistency across models. Extreme values in assets, expenditures, and land area are winsorized at the 99th percentile.

Table 1 defines the key variables used in the analysis, and Table 2 reports summary statistics for the full sample and by farm size quartile. The proportion of landholders receiving agricultural loans is uniformly low across the distribution of farm size, ranging from 4.1 percent of households in the bottom quartile to 6.2 percent in the top quartile. Mean loan size varies modestly, from ₦96.7 thousand in the bottom quartile to ₦108.8 thousand in the top quartile (a 12.4 percent difference). Notably, the second quartile reports the highest mean loan size (₦110.2 thousand), indicating that loan amounts do not increase monotonically with farm size. This is noteworthy given the relative large differences in landholdings, with the average farm size rising from 0.11 hectares in the bottom quartile to 3.16 hectares in the top quartile (an increase of more than 20-fold across quartiles).[3]

Expenditure patterns differ substantially across the farm size distribution. Households in the top quartile have considerably higher farm expenditures but lower nonfarm expenditures than those in the bottom quartile. Although expenditure profiles vary sharply with land endowments, access to credit remains limited across the distribution, consistent with broad credit rationing.

[2]Food expenditures are based on 7-day recall, while some agricultural expenditures use longer recall windows. All are converted to annual values.

[3]More details about the distribution of farm size by loan access can be found in Figure A2 of the supplementary Appendix A.

Table 1.
Description of Key Variables

Treatment Variable	**Description**
Received Agricultural Loan (Binary 0/1)	Indicator variable, true when the household received a loan with a stated purpose for farm expenditures.
Outcome Variables	
Input Expenditures (1,000s ₦)[a]	Seeds, fertilizer, pesticides, machinery/tools.
Land Expenditures (1,000s ₦)[a]	Land purchases and rent.
Labor Expenditures (1,000s ₦)[a]	Hired labor, rented livestock labor.
Livestock Expenditures (1,000s ₦)[a]	Livestock purchases, livestock upkeep costs.
Farm Expenditures (1,000s ₦)	Sum of: Input, Land, Labor, and Livestock Expenditures.
Food Expenditures (1,000s ₦)[a]	Imputed value of food consumption, the product of purchased food consumed and respondent recalled market price. Averaged between pre-planting and post-harvest surveys.
Non-Food Expenditures (1,000s ₦)[a]	Household expenditures on goods (durable and non-durable, e.g. cigarettes, appliances), and services (e.g. vehicle repairs, insurance). **Excludes:** nonfarm business, education, health, debt, rent, vehicles, ceremonies, and holidays.
Nonfarm Expenditures (1,000s ₦)	Sum of Food and Non-Food Expenditures: intended to represent average daily consumption while excluding lumpy irregular consumption events.
Select Control Variables	
Farming Land Area (Hectares)[a,b]	Size in hectares of land owned by household, include actively harvested, fallow fields, and pasture land.
Value of Farm Production (1,000s ₦)[a,b]	Value of all crop production produced by a household, constructed from production statistics multiplied by applicable crop prices by region.
Assets (1,000s ₦)[a,b]	Sum of the estimated value of durable goods owned by a household (excluding dwelling and agricultural plots).
Nonfarm Income (1,000s ₦)[a,b]	Income from nonfarm businesses, remittances, and all other sources.
Livestock Holdings (#)[b]	The Tropical Livestock Unit value for a household in the post-planting survey. Provides a standardized measure of livestock density including all recorded livestock such as cows, sheep, and chicken.
Has Formal Land Rights (Binary 0/1)[b]	Related to agricultural plots, the household has documentation of land rights (at least one plot).
Head Marital Status (Binary 0/1)	Indicates if the head of the household is currently married, all other classifications such as divorced, separated, or widowed are classed as zero.
Food Consumption Score (#)	The Food Consumption Score (FCS) is an index that measures the diversity and frequency of food groups consumed, which is then weighted according to the relative nutritional value of the consumed food groups (ranges 0 to 100).
Received Non-Agricultural Loan (Binary 0/1)	Indicator variable, true when the household received a loan with a stated purpose besides farm expenditures.

[a] Winsorized to top 1%
[b] Imputation performed by Evans School Policy Analysis and Research Group (EPAR, 2025).

Table 2.
Summary Statistics by Farm Size Quartile

	Mean	Mean by Farm Size Quartiles			
	Full Sample	Q1	Q2	Q3	Q4
Agricultural Loan Variables					
Received Agricultural Loan (Binary 0/1)	0.05	0.04	0.04	0.05	0.06
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	102.85	96.75	110.20	82.34	108.77
Outcome Variables					
Input Expenditures (1,000s ₦)[a]	33.20	14.55	30.58	39.37	55.25
Land Expenditures (1,000s ₦)[a]	4.29	2.63	4.32	4.38	6.32
Labor Expenditures (1,000s ₦)[a]	64.56	31.55	58.17	66.04	102.44
Livestock Expenditures (1,000s ₦)[a]	32.31	19.07	36.16	35.20	34.80
Farm Expenditures (1,000s ₦)	117.98	56.71	112.89	131.52	188.76
Food Expenditures (1,000s ₦)[a]	1,538.36	1,660.10	1,569.41	1,484.12	1,351.75
Non-Food Expenditures (1,000s ₦)[a]	427.67	445.44	424.95	402.94	414.76
Nonfarm Expenditures (1,000s ₦)	1,966.03	2,105.53	1,994.36	1,887.06	1,766.51
All Fertilizer Use (kg/ha)	0.75	0.86	1.18	0.74	0.39
Ratios					
Agricultural Loan to Total Farm Expenditures	87.2%	170.6%	97.6%	62.6%	57.6%
Agricultural Loan to Nonfarm Expenditures	5.2%	4.6%	5.5%	4.4%	6.2%
Control Variables					
Farming Land Area (Hectares)[a,b]	1.18	0.11	0.43	1.03	3.16
Value of Farm Production (1,000s ₦)[a,b]	355.75	88.01	250.43	405.45	679.10
Assets (1,000s ₦)[a,b]	276.68	316.97	259.25	236.84	280.37
Nonfarm Income (1,000s ₦)[a,b]	349.87	367.59	361.43	322.12	299.64
Livestock Holdings (#)[a,b]	0.79	0.21	0.45	0.82	1.56
Has Formal Land Rights (binary 0/1)[a,b]	0.14	0.11	0.12	0.14	0.19
Income Shock (#)	0.20	0.20	0.18	0.21	0.21
Production Shock (#)	0.95	1.00	0.99	0.97	0.89
Price Shock (#)	1.07	1.33	1.14	1.04	0.78
Head Marital Status (binary 0/1)[c]	0.77	0.60	0.78	0.83	0.89
Head Age[c]	51.74	55.56	51.17	49.86	49.96
Head Sex (binary $1 \equiv$ Male)[c]	0.81	0.65	0.83	0.86	0.92
Number of Members in the Household	5.93	4.48	5.81	6.33	7.21
Number of Adult Members in the Household	2.93	2.48	2.83	3.08	3.34
Household Has Phone Access (binary 0/1)	0.93	0.92	0.93	0.93	0.95
Household Has Internet Access (binary 0/1)	0.36	0.41	0.34	0.32	0.34
Probability of Moderate Food Insecurity	0.46	0.65	0.47	0.39	0.31
Food Consumption Score	49.02	48.01	48.20	48.92	50.14
Received Non-Agricultural Loan (binary 0/1)	0.09	0.10	0.09	0.07	0.07

[a] Winsorized to top 1%.
[b] Imputation performed by Evans School Policy Analysis and Research Group (EPAR, 2025).
[c] Characteristics of the Head of Household as noted on survey responses.
Note: We conducted ANOVA p-value tests of equality across quartile means and found all variables statistically significant (at least at the 10% level) except Income Shocks.

2.3 Loans, Economic Shocks, and Food Security Data

The LSMS household questionnaire conducted during the post-planting visit records loan applications in the twelve months preceding the survey, approvals, sources, and intended use[4]. Pooling data across waves, roughly one-third of loan applications were earmarked for farming, with approximately 76% of agricultural loans sourced from informal or semi-formal sources (e.g. friends, money-lenders, savings associations)[5].

To approximate the effect of agricultural credit programs (such as the Anchor Borrowers Program), we use loans specifically earmarked for farm expenditures as a proxy for agricultural credit[6]. We pool loans across informal and formal sources for two reasons: (i) both sources are effectively fungible in relaxing the household liquidity constraint, and (ii) the scarcity of formal agricultural loans necessitates pooling to preserve statistical power.

We define credit access at the extensive margin (a binary treatment variable), because relaxing binding budget constraints often induces structural changes in household expenditure behavior, such as 'crowd-in' effects (Banerjee et al., 2019). Furthermore, this approach mitigates endogeneity concerns regarding the correlation between loan magnitude and household assets. Additionally, to isolate the treatment effect and account for households with multiple loans, we include a control dummy variable for non-agricultural loans.

To control for exogenous disturbances, we use the Economic Shocks module within the household questionnaire during the post-harvest visit, which records unexpected events affecting the household's welfare in the three years preceding the survey. Because the specific shock categories vary slightly between waves (21 categories in Wave 4 and 28 in Wave 5), we harmonize them into four distinct categories: (1) Production Shocks (e.g. drought, floods, pests), (2) Income Shocks (e.g. death of earner, loss of remittances), (3) Price Shocks (e.g. fuel price inflation, input/output price volatility), (4) Other Shocks (e.g. conflict, theft). We restrict the shock window to the two years preceding the survey wave to reflect acute events which might induce or be ameliorated by credit acquisition. Specifically, we sum the number of shocks experienced by the household in each category to capture the acute intensity of shocks experienced by the household not absorbed by fixed effects.

Finally, we control for food security status. We measure household food security using

[4] The specific wording of the survey questionnaire is "In the last 12 months, have you or anyone else in the household attempted to borrow money or applied for or received a loan from [a list of plausible sources]?" There is no indication that loans were made "in-kind" and specific values in Naira are provided for each loan received.

[5] See Figure A3 in Appendix A for more details on the distribution of farm size by loan formality.

[6] The LSMS datasets do not include information on whether a loan originates from a specific government program, or if it otherwise could be formally defined as an agricultural loan.

the Food Consumption Score and the Food Insecurity Experience Scale (FIES), calculated using data from the Food Security module present in the household questionnaire post-harvest visit in both waves. For FIES, we follow Cafiero et al. (2018) and estimate the probability of a household experiencing either moderate or severe food insecurity.

3 Empirical Framework

3.1 Estimation Strategy

Our objective is to estimate how access to agricultural credit affects household expenditure allocation, and whether these effects vary with farm size. We exploit the panel structure of the LSMS data, using within-household variation between Wave 4 (2018–2019) and Wave 5 (2023–2024) to control for time-invariant unobserved heterogeneity.

A conventional two-way fixed effects (TWFE) model would address household-level fixed characteristics and common time shocks. However, given the high dimensionality of observable covariates and the potentially complex selection into credit, we adopt a Double/Debiased Machine Learning (DML) framework[7]. DML operates by partialling out the confounding effects of observables from both outcome and treatment variables, leveraging high-dimensional machine learning functions to isolate pseudo-random variation in credit access[8]. This approach preserves the fixed-effects structure while allowing flexible, non-parametric modeling of the relationships between covariates, treatment assignment, and outcomes.

In the following subsections, we outline the DML specification, its adaptation to panel data, and the implementation details of our estimation procedure.

3.2 Identification

3.2.1 Double/Debiased Machine Learning (DML) Specification

To isolate the causal effect of agricultural loan use on household expenditures while accounting for the high-dimensional confounders pervasive in credit acquisition, we employ a Double/Debiased Machine Learning (DML) approach. Unlike traditional fixed effects models, DML uses non-parametric modeling of the relationship between covariates,

[7]For our analysis, we primarily utilize the DoubleML package developed for Double Machine Learning in Python (Bach et al. , 2022).

[8]DML necessarily relies upon the assumption of conditional independence; as such, we explore the robustness of our results to omitted variable bias in Section 4.5.

treatment assignment, and outcomes, thereby improving identification when compared to traditional econometric techniques (Belloni et al., 2014; Chernozhukov et al., 2018, 2025; Semenova and Chernozhukov, 2020).

We estimate a Partial Linear Regression (PLR) model with an interaction term, defined as:

$$Y_{it} = \theta_1 L_{it} + \theta_2 \left(L_{it} \times \hat{A}_{it}\right) + g\left(X_{it}, \bar{X}_i\right) + u_{it} \tag{1}$$

where Y_{it} is the outcome and denotes each category of expenditure for household i at time t (Expenditures), L_{it} is the treatment and denotes whether or not the household received an agricultural loan (Received Agricultural Loan), and $\hat{A}_{it}$ represents the centered farm size in hectares (Farming Land Area) defined as:

$$\hat{A}_{it} = A_{it} - \bar{A}, \tag{2}$$

where $\bar{A}$ is the overall sample mean across all households i and survey waves t. In equation 1, the unobserved, non-parametric function $g(\cdot)$ captures confounding variance for each category of expenditure separately through both time-varying controls X_{it} and their Mundlak means $\bar{X}_i$, calculated for each household over time, controlling for time-invariant heterogeneity. We first estimate the model without the interaction term to obtain the baseline treatment effect, and then include the interaction term to test for heterogeneity based on farm size. Following Chernozhukov et al. (2025), we treat the interaction term $\left(L_{it} \times \hat{A}_{it}\right)$ as a second conditional average treatment effect such that $\hat{A}_{it} \in X_{it}$. This allows us to jointly estimate the baseline average treatment effect θ_1, and the heterogeneous treatment effect θ_2. The equations are estimated separately for each category of farm or nonfarm expenditures, and then for each of the sub-categories in farm expenditures (input, land, labor, livestock expenditures) and nonfarm expenditures (food and non-food expenditures).

3.2.2 DML in Panel Data

Adapting DML to a panel dataset requires careful handling of unobserved heterogeneity. Traditional DML assumes independence of observations, which is violated by repeated household observations as is the case here. Additionally, producing TWFE inference from DML is challenging and demeaning data prior to processing can introduce bias. To address these challenges, we implement two modifications aligned with the best practices emerging from the newly expanding DML panel data literature (Fuhr and Papies, 2024; Clarke and Polselli, 2025).

First, we incorporate Correlated Random Effects (CRE) by augmenting the control matrix X_{it} with household-level time means (Mundlak means) of all time-varying controls ($\bar{X}_i$). This allows us to account for time-invariant unobserved heterogeneity, emulating a two-way fixed effects specification which would otherwise be biased in the DML framework (Clarke and Polselli, 2025). Additionally, we include a time dummy for the second time period (second wave of the survey) inside X_{it} to account for aggregate shocks. Taken together, these modifications allow DML to produce results analogous to a traditional TWFE model, but with the non-parametric flexibility of DML.

Second, we enforce grouped cross-fitting[9]. Specifically, the cross-fitting procedure relies on the independence of observations when dividing data into groups ("folds"), to address within households inter-temporal correlation we group data at the household level. As a robustness check, we explore the sensitivity of our results to different cross-fitting specifications in Section 4.4.

3.2.3 Stacked Learners and Rich Control

In the DML framework, the conditional expectation of the outcomes $\hat{E}[Y|X]$ and the treatment $\hat{E}[L|X]$ serve as nuisance function parameters. Specifically, functions that must be estimated to partial out confounding but are not the core objective of the analysis. Therefore, the validity of our causal estimates hinge on predictive accuracy of our machine learning models used to approximate these functions. Consequently, to maximize model performance, we implement a specific training and estimation strategy.

First, regarding sample composition: due to a logarithmic transformation on outcomes Y, we exclude household observations with zero values for their expenditures from our models resulting in smaller sample sizes for sub-category analysis. Additionally, to maintain consistency across model specifications and maximize first-stage accuracy, we estimate $\hat{E}[L|X]$ once on the full paired panel ($N = 6,398$ pooled observations) and reuse it across outcome-specific models. We employ median imputation with missingness indicators to leverage the complete information set. However, the outcome nuisance function and final second-stage estimation of treatment effects are restricted to the subsam-

[9]Cross-fitting is a common technique utilized in machine learning models to avoid overfitting the model. First the dataset is divided into subsets ("folds"), where a model is trained on one group of folds, and predictions are made on another. This reduces the risk of fitting a model to noise and introducing bias. Importantly, this methodology assumes observations in different folds are independent.

ple ($N = 5,900$ pooled observations) with non-missing values for the interaction variable (farm size). This strategy allows our nuisance function to benefit from maximum data depth while ensuring a consistent nuisance function estimation across the baseline and heterogeneous model specifications. Specifically, outcome nuisance functions ($\hat{E}[Y|X]$) are implemented on each outcome specific sample. Our strategy of training the treatment nuisance function on the maximal available sample prior to subgroup evaluation directly follows the Group ATE framework in Chernozhukov et al. (2025).

Second, we employ a "stacked" ensemble for our learners by training several different types of models and a rich set of covariates in X_{it}. We use ensemble learners as a best practice technique for nuisance function estimation to improve the stability and generalization of our results. Our ensemble integrates a diverse set of algorithms to balance bias and variance: (i) Tree Based Models, including Random Forest and Histogram Gradient Boosting to capture non-parametric relationships, and (ii) Regularized Linear Models, including Lasso, Ridge, and Elastic Net to handle high-dimensionality and overfitting.

We use distinct learners for our outcome and treatment variables. For the binary agricultural loan treatment, we use classification variants of the learners (e.g. Logistic Regression with Elastic Net penalty, and stacking classifier), while we employ their regression counterparts for the continuous expenditures outcome. The meta learner (final learner) for each ensemble are Ridge model for the continuous outcome and a Logit for the binary treatment. Finally, to compute the interaction term (technical treatment), the predicted agricultural loan treatment variable was interacted with the farm size.

Complementing this robust learner set, X_{it} comprises a rich vector of socio-economic and fixed effect variables. As we mentioned above, to account for time-varying heterogeneity, we calculate Mundlak means for all controls and cluster at the household level. Additionally, we include state and time fixed effects, corresponding to the Nigerian Federal states and the wave indicator variables, respectively. Finally, we address covariates' missingness[10] in the nuisance stage by using median imputation with missingness indicators, consistent with the missingness indicator approach in Chernozhukov et al. (2025). To avoid leakage, median imputation is fit within each training fold.

3.2.4 Inference and Robustness

To ensure the stability of our results, we repeat the cross-fitting procedure over 30 random repetitions of fold selection for cross-fitting and aggregate results using the median

[10] 612 of 5,900 paired observations (10.4%) have at least one missing value among the control variables in our dataset, excluding observations with missing farm size values.

(Chernozhukov et al., 2018, 2025). We assess predictive validity of the nuisance functions using three primary diagnostics. For the outcome model, we report a standard R^2 (R^2_Y) value. For the treatment propensity model, we report the Area Under the Curve (AUC) and Average Precision (AP). We select these complementary metrics to address the specific challenges of our data: AUC[11] provides a standard measure of discrimination often used in traditional econometrics, while AP[12] offers a superior benchmark for unbalanced treatment and is commonly used in machine learning model assessment. Ideal metrics indicate that the learners are identifying genuine signal without overfitting to noise. Finally, to ensure our estimates are robust to fold specification, we test the robustness of our baseline 5-fold estimates against alternative 3- and 8-fold specifications.

4 Results

This section presents the empirical results from the DML models. We interpret the estimates as reflecting changes in household expenditure allocation following the receipt of agricultural credit. Because credit relaxes liquidity constraints at the household level, observed expenditure responses capture reallocation across farm and nonfarm uses rather than adherence to stated loan purposes. We first report baseline average treatment effects, then examine heterogeneity by farm size, followed by disaggregated expenditure analyses and robustness checks.[13]

4.1 Baseline results

Table 3 reports the baseline average treatment effects (ATE) of receiving an agricultural loan on household expenditures. Because the outcomes are log-transformed and treatment is binary, coefficients are interpreted as percentage changes using $100 \times (e^{\beta} - 1)$. Observations with zero expenditures are excluded, so estimates reflect intensive-margin responses for households with positive expenditures.

Receiving an agricultural loan increases both farm and nonfarm expenditures by 14.5

[11] The Area Under the Curve measures the probability a model correctly ranks a randomly chosen treated observation higher than a randomly chosen untreated observation. A value of 0.5 indicates predictive power equivalent to random choice, while a value greater than 0.5 indicates the model assigns treatment probabilities more accurately than random chance.

[12] Average Precision summarizes a model's precision recall curve over all decision thresholds ($p(treatment|X) \geq \tau,\ \tau \in [0,1]$). A naive model that assigns treatment at random will yield an AP equal to the sample prevalence of treatment. Therefore, AP is beneficial to contextualize the model's predictive power in settings such as ours where treatment is unbalanced and rare.

[13] A balance table and additional descriptive figures are reported in Appendix A Table A1.

percent, statistically significant at the 10 percent and 1 percent levels, respectively. These estimates exploit within-household variation across survey waves and control for observed covariates, household correlated random effects, and common time shocks.

Table 3.
Average Treatment Effects of Agricultural Loans on Expenditures

	Farm Expenditures (log)	Nonfarm Expenditures (log)
Received Agricultural Loan	0.136*	0.135***
	(0.071)	(0.049)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household CREs	Yes	Yes
Learners		
Loan $\sim X$	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R
K-folds	5	5
Diagnostics		
R^2_Y (Outcome: $Y \sim X$)	0.351	0.420
AUC (Loan $\sim X$)	0.626	0.630
AP (Loan $\sim X$)	0.083	0.082
Observations	5,476	5,900

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$
R^2_Y report nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance.
performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

These results are consistent with experimental evidence showing that relaxing an agricultural household's budget constraint raises both agricultural investment and nonfarm spending. For example, Delavallade and Goldlonton (2023) show that warrantage (inventory credit) in Burkina Faso increases total per-capita expenditures by 13% (driven by non-food categories) and raises crop revenues by roughly 15%. Similarly, Beaman et al. (2023) find sizable expenditure responses to liquidity provision in Mali.

The predictive validity of our nuisance functions is substantiated by the Area Under the Curve (AUC) and Average Precision (AP) statistics reported in Table 3. The AUC is a common diagnostic statistic for binary outcome models, and can generally be interpreted as the probability that a random positive instance is ranked higher than a negative one. Our AUC values of 0.626 and 0.630 (farm and nonfarm expenditures models, respectively) structurally outperform the naive (random) assignment model benchmark of 0.5. While these scores suggest relatively poor discrimination by traditional standards (Hosmer et al., 2013), they align with recent literature on the difficulty of predicting credit behavior in opaque and noisy smallholder microdata. For instance, Björkegren and Grissen (2020) demonstrate that machine learning models constrained to traditional demographic data often yield AUC scores of between 0.51 and 0.57 for credit repayment outcomes.

The AP is a standard machine learning metric that summarizes the precision-recall curve. It can generally be interpreted as the average probability that a positive prediction made by the model is correct, benchmarked against the baseline probability of random selection (sample prevalence). Our AP values of 0.083 and 0.082 (farm and nonfarm models, respectively) indicate moderate discrimination above random assignment. Given the low prevalence of agricultural credit in our sample (4.9%), a naive model would achieve an AP of only 0.049. Thus our models achieve a precision roughly 66.3–69.6% higher than a naive benchmark. Collectively, these diagnostics indicate that despite the opaque nature of informal credit markets where soft variables (e.g. trust, reputation) heavily influence credit access, our learners successfully isolate systemic selection signals necessary for DML nuisance function estimation.

4.2 Heterogeneous treatment effects by farm size

We next examine whether the impact of agricultural credit varies with farm size. Table 4 augments the baseline specification with an interaction between loan receipt and farm land area (centered at the sample mean). To supplement our analysis, we also conduct a quartile regression analysis, more broadly referred to as a Group Average Treatment Effect (GATE) analysis in DML, in Appendix B.

The baseline coefficient on loan receipt remains positive and statistically significant for farm expenditures (0.136 log points, approximately 14.5 percent). The coefficient on the interaction term is also positive (0.079) and statistically significant, indicating that the farm expenditure response increases by 8.2 percentage points for each additional hectare of land. In contrast, the interaction term for nonfarm expenditures is small (–0.021) and statistically insignificant, suggesting that the estimated 14.5 percent increase (0.135 log points) in nonfarm expenditures does not vary systematically with farm size.

Table 4.
Heterogeneous Effects of Agricultural Loans on Expenditures

	Farm Expenditures (log)	Nonfarm Expenditures (log)
Received Agricultural Loan (Loan)	0.136*	0.135***
	(0.070)	(0.049)
Loan $\times$ Farming Land Area (HA)	0.079*	-0.021
	(0.045)	(0.031)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household CREs	Yes	Yes
Learners		
Loan $\sim X$	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R
K-folds	5	5
Diagnostics		
R^2_Y (Outcome: $Y \sim X$)	0.351	0.420
AUC (Loan $\sim X$)	0.626	0.630
AP (Loan $\sim X$)	0.083	0.082
Observations	5,476	5,900

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$
R^2_Y report nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance.
performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To identify the structural tipping point between farm investment and consumption smoothing, we equate the two marginal treatment effects. Because the interaction term is centered at the mean farm size across waves (1.22 ha), the marginal effect of agricultural credit on farm expenditures is given by the baseline coefficient plus the interaction effect scaled by the deviation from the mean. We solve for the farm size tipping point (A^*), where the increasing farm investment effect intersects the constant nonfarm consumption effect:

$$\underbrace{0.135}_{\text{Nonfarm Effect}} = \underbrace{0.136 + 0.079 \times (A^* - 1.22)}_{\text{Farm Effect}}$$

Solving for A^* yields a threshold of approximately 1.20 hectares. This corresponds to approximately the 67^{th} percentile of the farm size distribution[14]. Below this threshold, agricultural credit induces a greater marginal effect on consumption expenditures; above it, the marginal effect of credit on farm investment dominates.

These results highlight structural heterogeneity in the constraints facing smallholders. Specifically, farms below the threshold of 1.20 hectares are likely facing binding consumption constraints, necessitating the allocation of liquidity from agricultural credit towards immediate subsistence needs. Conversely, farms operating above this threshold face less binding consumption needs and can allocate credit toward agricultural intensification. This divergence suggests that landholdings and agricultural credit function as structural complements, enabling larger smallholders to realize the long-term benefits of agricultural credit access.

Importantly, these heterogeneous effects are unlikely to be driven by differences in loan magnitude. As shown in Table 2, the ratio of agricultural loans to farm expenditures is higher among the smallest farms (170.6 percent in Quartile 1) than among the largest farms (57.6 percent in Quartile 4). Despite receiving loans that are large relative to their existing farm expenditures, smaller farms exhibit weaker farm investment responses. This pattern suggests that the observed heterogeneity reflects differences in expenditure allocation rather than differences in credit volume.

4.3 Sub-category analysis for farm and nonfarm expenditures

To further examine the specific mechanisms driving the effect of agricultural loans on household expenditures we perform analysis based on the components of farm expenditures (Inputs, Land, Labor, and Livestock) and nonfarm expenditures (Food, and Nonfood).

Table 5 presents the disaggregated farm expenditure treatment effects. For input, land, and labor expenditures there are no statistically significant results. This is likely a result of heterogeneity in investment strategies across households (i.e. some households intensify through input expenditures such as fertilizer utilization, while others hire labor or acquire land) and reduced power due to smaller sample sizes. However, coefficient signs are universally positive and for livestock expenditures, receiving agricultural loans increases livestock expenditures by 46.3% (0.382 log points) at a level of significance of 10%. Given

[14] This threshold is calculated using unrounded point estimates and the mean farm size specific to the farm expenditure subsample ($N = 5,476$).

that this category includes recurring costs such as feed and healthcare, this pattern is consistent with credit supporting ongoing capital maintenance rather than extensive margin herd expansion. However, the null result on disaggregated input expenditures (Column 1) may obscure specific treatment effects.

Table 5.
Heterogeneous Effects of Agricultural Loans on Disaggregated Farm Expenditures

	Input Exp. (log)	Land Exp. (log)	Labor Exp. (log)	Livestock Exp. (log)	Fertilizer (log kg/ha)
	(1)	(2)	(3)	(4)	(5)
Received Agricultural Loan (Loan)	0.048	0.093	0.064	0.382^{*}	0.015
	(0.097)	(0.211)	(0.081)	(0.198)	(0.116)
Loan × Farming land area (HA)	0.068	0.044	0.044	0.054	0.186^{***}
	(0.061)	(0.176)	(0.048)	(0.086)	(0.069)
Specification					
Controls	Yes	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes	Yes	Yes
Learners					
Loan $\sim X$	PLH-L	PLH-L	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R	ERH-R	ERH-R
K-folds	5	5	5	5	5
Diagnostics					
R^2_Y (Outcome: $Y \sim X$)	0.362	0.234	0.238	0.226	0.386
AUC (Loan $\sim X$)	0.605	0.661	0.620	0.633	0.599
AP (Loan $\sim X$)	0.079	0.117	0.089	0.097	0.085
Observations	3,670	352	4,010	1,176	1,676

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$

R^2_Y report nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance..

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To test whether the size-dependent investment effects observed in our heterogeneous effects model (Table 4) are driven by agricultural intensification, we specifically examine fertilizer application intensity (kg/ha) in Column 5. The results suggest that fertilizer intensification is important for larger smallholders who receive credit. The coefficient (0.186 log points) on the interaction term is positive and statistically significant at the 1% level,

Table 6.
Heterogeneous Effects of Agricultural Loans on Nonfarm Expenditures

	Food Expenditures (log)	Non-Food Expenditures (log)
Received Agricultural Loan (Loan)	0.146**	0.069**
	(0.060)	(0.035)
Loan × Farming land area (HA)	−0.043	0.036
	(0.039)	(0.026)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household CREs	Yes	Yes
Learners		
Loan $\sim X$	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R
K-folds	5	5
Diagnostics		
R^2_Y (Outcome: $Y \sim X$)	0.335	0.604
AUC (Loan $\sim X$)	0.630	0.630
AP (Loan $\sim X$)	0.082	0.082
Observations	5,898	5,900

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$
R^2_Y report nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance.
performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

indicating a HTE of receiving an agricultural loan on fertilizer utilization of 20.4% per hectare. Of particular note, this positive estimate contrasts with the descriptive gradient we observe in Table 2, where the smallest farms utilize fertilizer more intensively (Quartile 1: 0.86 kg/ha) than the largest farms (Quartile 4: 0.39 kg/ha). This suggests that when scale permits profitable intensification, agricultural credit can translate into productivity-enhancing input use, reinforcing the heterogeneity among smallholders and the importance of aligning credit design with farm characteristics.

Table 6 presents treatment effects for the components of nonfarm expenditures. Receiving an agricultural loan increases food expenditures by 15.8 percent (0.146 log points) and non-food expenditures by 7.1 percent (0.069 log points), both statistically significant at the 5 percent level. The interaction terms with farm size are statistically insignificant, consistent with the earlier finding that nonfarm expenditure responses do not vary systematically with landholdings.

Although the estimated effects on food expenditures are sizable, they should not be interpreted as a one-to-one mapping from loan value to consumption. As shown in Table 2, the ratio of median agricultural loan value to total food expenditures is relatively small (4.4–6.2 percent across quartiles). The estimated percentage increases therefore likely reflect broader household reallocation following liquidity relief, rather than direct spending of loan proceeds. In this sense, credit access appears to relax short-run budget constraints, allowing households to adjust consumption patterns alongside farm investment decisions.

4.4 Sensitivity to cross-fitting folds

We assess the stability of our core estimates to model specification. Specifically, DML results can be sensitive to the random partitioning of data during the cross-fitting process. Tables 7 and 8 demonstrate the sensitivity of our findings to model design by varying the number of cross-fitting folds ($K \in 3, 5, 8$). The results indicate that our estimates are highly stable across fold specification. The greatest sensitivity is observed in Table 8, with a difference of approximately 0.005 between the 3-fold and 8-fold specifications. The majority of point estimates differ by less than 0.003 across fold specifications.

Table 7.
Baseline Sensitivity Analysis for Farm and Nonfarm Expenditures

	3 Folds	5 Folds	8 Folds
Panel A: Log Farm Expenditures			
Received Agricultural Loan	0.137*	0.136*	0.135*
	(0.072)	(0.071)	(0.071)
R^2_Y (Outcome: $Y \sim X$)	0.346	0.351	0.352
AUC (Loan)	0.626	0.626	0.622
AP (Loan)	0.083	0.083	0.082
Observations	5,476	5,476	5,476
Panel B: Log Nonfarm Expenditures			
Received Agricultural Loan	0.137***	0.135***	0.132***
	(0.049)	(0.049)	(0.048)
R^2_Y (Outcome: $Y \sim X$)	0.417	0.420	0.422
AUC (Loan)	0.622	0.630	0.625
AP (Loan)	0.081	0.082	0.080
Observations	5,900	5,900	5,900
Specification			
Controls	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes
Learners			
Loan $\sim X$	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.
R^2_Y report nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8.
Heterogeneous Sensitivity Analysis for Farm and Nonfarm Expenditures

	3 Folds	5 Folds	8 Folds
Panel A: Log Farm Expenditures			
Received Agricultural Loan	0.137^{*}	0.136^{*}	0.135^{*}
	(0.072)	(0.071)	(0.071)
Loan × Farming Land Area (HA)	0.081^{*}	0.079^{*}	0.080^{*}
	(0.045)	(0.045)	(0.045)
R_Y^2 (Outcome: $Y \sim X$)	0.346	0.351	0.352
AUC (Loan)	0.619	0.626	0.622
AP (Loan)	0.083	0.083	0.082
Observations	5,476	5,476	5,476
Panel B: Log Nonfarm Expenditures			
Received Agricultural Loan	0.137^{***}	0.135^{***}	0.132^{***}
	(0.049)	(0.049)	(0.048)
Loan × Farming Land Area (HA)	−0.022	−0.021	−0.020
	(0.031)	(0.031)	(0.031)
R_Y^2 (Outcome: $Y \sim X$)	0.417	0.420	0.422
AUC (Loan)	0.622	0.630	0.625
AP (Loan)	0.081	0.082	0.080
Observations	5,900	5,900	5,900
Specification			
Controls	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes
Learners			
Loan $\sim X$	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.
R_Y^2 report nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Robustness to Omitted Variable Bias

Approximately 76% of respondents in our sample stated that credit originated from informal sources (e.g. friends, money-lenders, savings associations). As a result, selection into credit is tied to opaque social factors. Consequently, unobservable confounders such as reputation, social networks, and entrepreneurial spirit represent a challenge to identification. This motivated our use of DML augmented for panel data, controlling for time-invariant unobservables with Mundlak means, and complex nonparametric relationships with our learners. The former design choice has been observed to significantly enhance performance over traditional linear TWFE specifications (Fuhr and Papies, 2024), while the latter has proven specifically useful in the context of noisy LMIC credit markets Delavallade and Goldlonton (2023).

Nevertheless, to address the threat of time-varying omitted variable bias (OVB) we perform a sensitivity analysis of our results following Chernozhukov et al. (2022). Specifically, we benchmark our treatment effect estimates against pseudo-OVB estimated on observables. We do this by sequentially removing observables from our control vector X, measuring the impact on the predictive power of our nuisance functions, and estimating a maximally adversarial (i.e. perfectly correlating confounding between treatment and outcome) confidence bound.

Figure 1 illustrates the confidence bounds of our point estimates over the six most impactful omitted variable combinations. We focus our discussion around the benchmarked impact of removing our core endowment variables: nonfarm income, household assets, and crop production value. Removing these principal endowment variables shifts our average treatment effects toward zero by up to 44.4% for consumption expenditures, 50.0% for farm expenditures, and 36.3% for the farm expenditures interaction term.

This bounding exercise indicates that an unobserved confounder would need to be approximately 2 to 2.8 times more impactful on both credit access and expenditures than the combined effect of our core endowment variables to drive our point estimates to zero. More conservatively, an unobserved confounder would need to exert roughly the same magnitude of influence as our core endowment variables to render our nonfarm expenditure ATE point estimates statistically insignificant at the 10% level, and about one-quarter the magnitude to do the same for our farm expenditure point estimate.

While soft variables such as reputation play an important role in accessing credit in informal settings, the magnitude of time-variant confounding necessary to nullify our results appears implausibly large. Regarding entrepreneurial drive, household expenditures re-

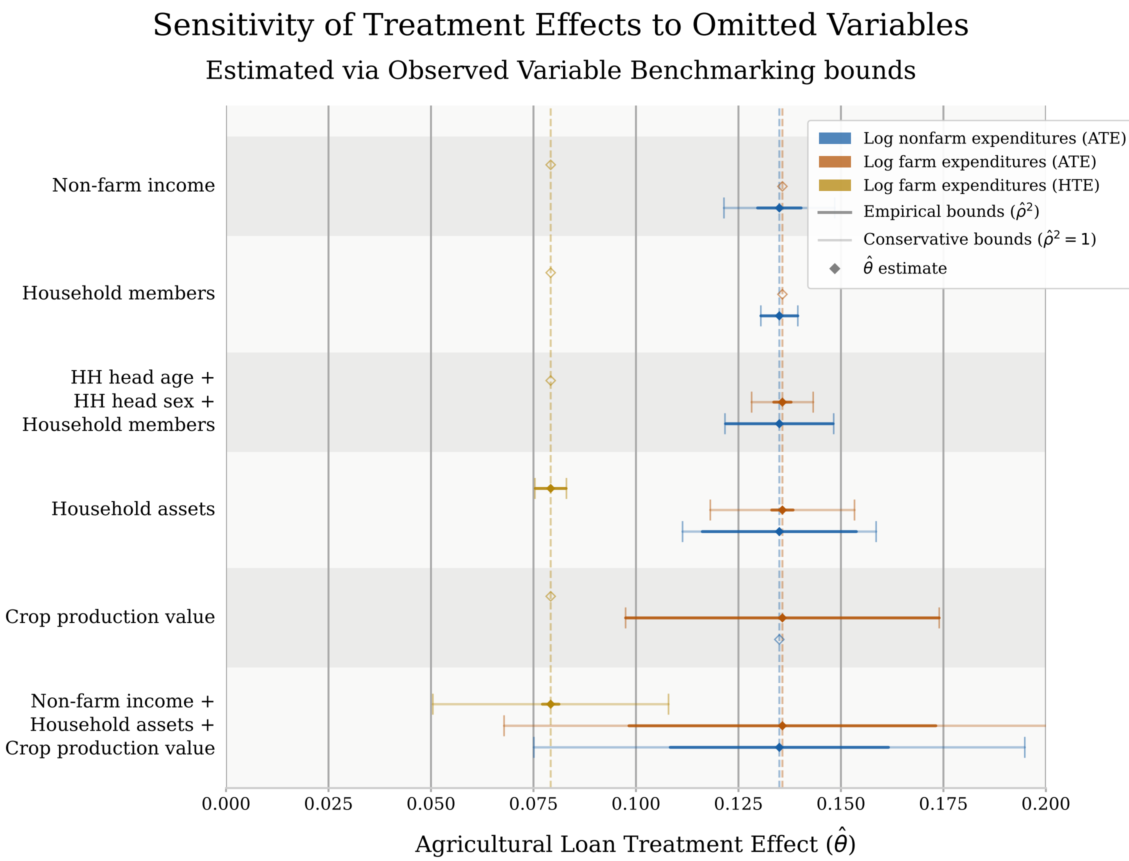
Sensitivity of Treatment Effects to Omitted Variables
Estimated via Observed Variable Benchmarking bounds
Log nonfarm expenditures (ATE)
Log farm expenditures (ATE)
Log farm expenditures (HTE)
Empirical bounds ($\hat{\rho}^2$)
Conservative bounds ($\hat{\rho}^2 = 1$)
$\hat{\theta}$ estimate
Non-farm income
Household members
HH head age +
HH head sex +
Household members
Household assets
Crop production value
Non-farm income +
Household assets +
Crop production value
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Agricultural Loan Treatment Effect ($\hat{\theta}$)

FIGURE 1

main fundamentally bounded by available assets and liquidity. Because soft variables are unlikely to dictate aggregate spending to the degree of fundamental economic indicators, we conclude that our findings are robust to unobserved confounding.

5 Discussion and Limitations

This paper examines how agricultural credit shapes household expenditure allocation in Nigeria and how these responses vary with farm scale. The results indicate that credit affects both farm and nonfarm expenditures, with heterogeneous effects that depend systematically on landholdings. These patterns align with models of production under incomplete markets, where the returns to purchased inputs depend on scale and complementary constraints (Carter and Olinto, 2003; Feder, 1985).

For relatively larger smallholders, agricultural credit is associated with stronger farm investment responses, including increased fertilizer intensity. This suggests that when scale permits profitable intensification, credit can facilitate productivity-enhancing input use, showing that credit and landholdings act like structural complements. In contrast, smaller farms exhibit comparatively weaker farm expenditure responses and stronger nonfarm adjustments, consistent with tighter consumption constraints and lower marginal returns to input intensification. Our results show that the smallest farms prioritize consumption smoothing over agricultural investment when receiving credit. These findings reinforce evidence from the broader agricultural finance and microcredit literature that credit impacts are highly heterogeneous and shaped by underlying production opportunities and risk environments (Karlan et al., 2014; Hussain and Thapa, 2016; Meager , 2019).

From a food systems perspective, the results highlight the dual role of agricultural credit. For some households, credit supports productive investment and potential yield gains; for others, it primarily eases short-run liquidity constraints. Evaluating agricultural finance policies solely on average production responses may therefore obscure important welfare and resilience effects. Designing credit instruments that account for farm heterogeneity may improve alignment between policy objectives and household behavior.

Several limitations warrant consideration. First, although the DML framework controls for high-dimensional observed confounding and incorporates correlated random effects to control for time-invariant unobserved confounding, identification relies on the assumption of conditional unconfoundedness. This is particularly relevant in settings where informal lending dominates and selection may depend on unobserved factors such as reputation or social networks. While the sensitivity analysis in Section 4.5 supports the robustness of our results to unobserved confounding, our identification strategy ultimately relies upon observables and cannot be held to the same causal certainty of an experimental setting. Second, loan magnitudes are weakly correlated with land endowments in our sample, suggesting that credit constraints remain binding across the farm size distribution. Consequently, the results reflect a context of pervasive credit rationing and may not generalize to environments with deeper formal financial penetration.

An additional limitation concerns measurement of food expenditures. Our food expenditure data is constructed from annualized 7-day recall data, which may introduce aggregation error and inflate expenditure levels relative to observed income streams. While our analysis focuses on percentage changes rather than levels, any systematic measurement error in food reporting could affect the magnitude of estimated effects. We therefore interpret the food expenditure results as indicative of directional responses rather than precise level shifts.

Finally, the estimated percentage effects on food expenditures are large relative to median loan size. This pattern suggests that credit access may induce broader expenditure reallocation rather than direct mechanical spending of loan proceeds. However, the data do not permit direct observation of intra-household allocation of loan funds, and we therefore interpret these responses as equilibrium adjustments following liquidity relaxation rather than as precise mappings from credit to specific expenditures.

6 Conclusions

This paper investigates how agricultural credit affects expenditure allocation among smallholder farmers in Nigeria and how these effects vary with farm scale. Using panel-adjusted Double Machine Learning, we estimate both average and heterogeneous treatment effects while accounting for high-dimensional confounding and time-invariant heterogeneity (Chernozhukov et al., 2018, 2025; Semenova and Chernozhukov, 2020).

Our findings indicate that agricultural credit influences both farm and nonfarm expenditures, with responses that depend systematically on landholdings. For relatively larger smallholders, credit is associated with stronger farm investment responses, including increased fertilizer intensity. For smaller farms, expenditure adjustments are more concentrated in nonfarm categories. These results are consistent with models of agricultural production under incomplete markets, where the profitability of purchased inputs depends on scale, supervision, and complementary constraints (Carter and Olinto, 2003; Feder, 1985). They also align with evidence from the broader microcredit and agricultural finance literature showing that credit functions as part of an integrated household portfolio rather than as project-specific capital (Angelucci et al., 2015; Banerjee et al., 2015; Meager , 2019).

By documenting continuous heterogeneity across the farm size distribution, this study helps reconcile mixed findings in prior work. Experimental and quasi-experimental studies have found positive investment responses when complementary conditions—such as insurance or favorable timing—are present (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2018). In settings characterized by pervasive credit rationing and exposure to shocks, however, liquidity relief may primarily facilitate consumption smoothing or risk management. Our Nigerian case reflects this duality, by showing that credit also acts as a complementary condition for agricultural investment but only for the larger smallholders.

Several caveats remain. Identification relies on the assumption of conditional unconfoundedness given a rich control vector and correlated random effects structure. Although the DML framework mitigates overfitting and high-dimensional bias (Belloni et al., 2014; Chernozhukov et al., 2018), informal lending environments may involve unobserved selection factors such as reputation or social ties. In addition, loan magnitudes are only weakly correlated with land endowments in our data, suggesting that credit constraints bind across the farm size distribution. The results therefore characterize behavior under severe credit rationing and may not generalize to contexts with deeper formal financial markets.

More broadly, the evidence underscores that agricultural credit serves multiple functions within rural households. It may support productive intensification where scale permits, while simultaneously enhancing resilience and consumption stability among more constrained farms. Designing financial instruments that recognize this heterogeneity, rather than targeting uniform production responses, may be central to advancing effective and equitable food system policy in low-income settings.

6.1 Policy and Methodological Implications

The findings have several implications for the design and evaluation of agricultural credit programs. First, the heterogeneous expenditure responses suggest that uniform loan products are unlikely to generate homogeneous outcomes. Credit appears to support productive intensification primarily among relatively larger smallholders, while functioning as short-run liquidity support among smaller farms. Credit design that accounts for farm scale (through differentiated loan sizes, repayment structures, or complementary input-linked arrangements) may better align financial instruments with production capacity and household constraints.

Second, the results are directly relevant to large-scale public lending initiatives in Nigeria, including NIRSAL and the Anchor Borrowers' Programme (ABP). These programs operate in environments characterized by severe credit rationing and heterogeneous farm structures. Our evidence suggests that scale-sensitive guarantees, repayment structures, and targeting mechanisms may improve program effectiveness. For smaller farms, credit may primarily enhance resilience and consumption stability; for relatively larger smallholders, it may facilitate input intensification and yield growth. Recognizing these differentiated channels may help refine eligibility criteria, loan terms, and complementary services within such programs.

Third, the findings highlight the importance of integrating credit with complementary institutions. In settings where risk exposure and input access constraints remain binding, liquidity alone may not translate into sustained productivity gains among smaller farms. Prior experimental evidence emphasizes the role of insurance, savings mechanisms, and seasonal timing in shaping investment responses (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2018). Aligning agricultural credit programs with risk-sharing mechanisms and input markets may therefore enhance their impact.

Methodologically, this study illustrates how machine-learning-based causal methods can inform agricultural finance policy. The Double Machine Learning framework enables estimation of heterogeneous treatment effects in high-dimensional observational settings (Chernozhukov et al., 2018, 2025). In contexts where randomized experiments are limited or politically costly, such tools can complement traditional evaluation strategies by identifying differential responses across household types and production scales.

More broadly, the evidence underscores that agricultural credit serves multiple functions within rural economies. Evaluations focused solely on average production effects risk overlooking welfare and resilience gains among households for whom credit primarily relaxes liquidity constraints. Designing financial architectures that acknowledge this heterogeneity is central to advancing effective and sustainable food system policy.

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Appendix A

TABLE A1.
Balance Table

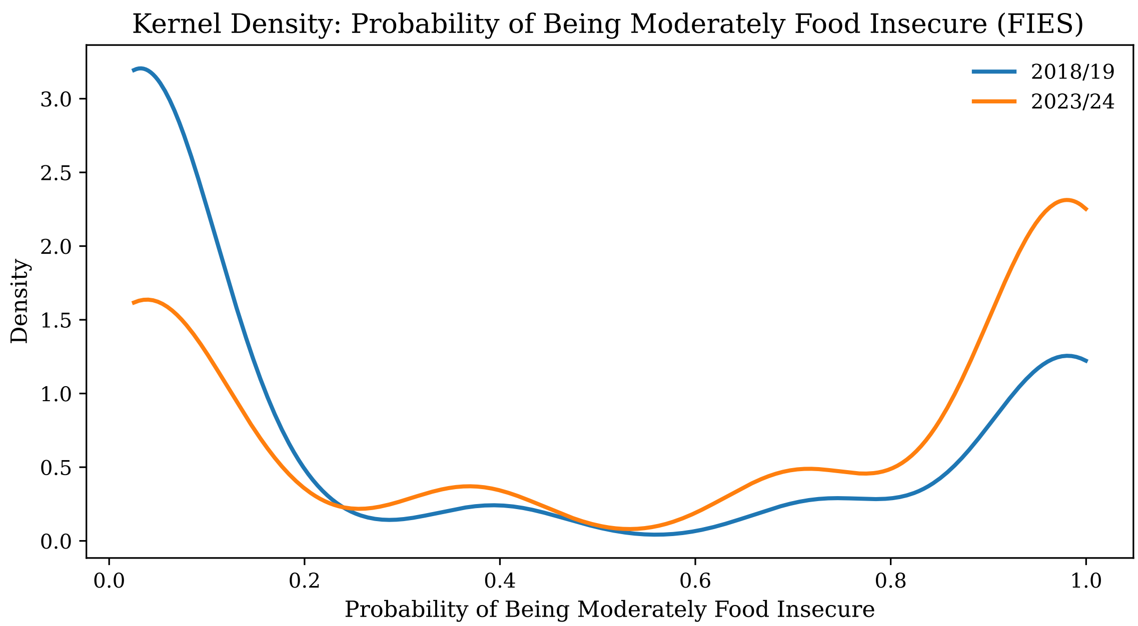
Variable	Did Not Receive Agricultural Loan (N ≈ 5,604) Mean (SE)	Received Agricultural Loan (N ≈ 296) Mean (SE)	Difference (1) - (2) Diff.
Farming Land Area (Hectares)	1.184 (0.019)	1.345 (0.085)	-0.161*
Value of Farm Production (1,000s ₦)	354.15 (7.61)	445.92 (36.78)	-91.768**
Assets (1,000s ₦)	270.21 (7.19)	228.16 (26.14)	42.048
Nonfarm Income (1,000s ₦)	330.09 (7.67)	377.97 (39.69)	-47.879
Livestock Holdings (#)	0.772 (0.024)	0.672 (0.084)	0.100
Has Formal Land Rights (binary 0/1)	0.137 (0.005)	0.193 (0.024)	-0.056**
Production Shock (#)	0.200 (0.007)	0.176 (0.026)	0.025
Food Shock (#)	0.962 (0.016)	1.078 (0.071)	-0.116
Price Shock (#)	1.069 (0.018)	1.108 (0.083)	-0.039
Head Marital Status (binary 0/1)	0.776 (0.006)	0.817 (0.023)	-0.041*
Head Age	51.745 (0.199)	48.990 (0.813)	2.755***
Head Sex (binary 1 ≡ Male)	0.815 (0.005)	0.845 (0.021)	-0.029
Number of Members in the Household	5.940 (0.046)	6.409 (0.184)	-0.469**
Number of Adult Members in the Household	2.932 (0.020)	2.959 (0.083)	-0.027
Household Has Phone Access (binary 0/1)	0.931 (0.003)	0.943 (0.014)	-0.011
Household Has Internet Access (binary 0/1)	0.349 (0.006)	0.365 (0.028)	-0.016
Probability of Moderate Food Insecurity	0.456 (0.006)	0.442 (0.025)	0.013
Food Consumption Score	48.618 (0.234)	50.243 (1.087)	-1.626
Received Non-Agricultural Loan (binary 0/1)	0.083 (0.004)	0.078 (0.016)	0.006

Notes: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Includes all observations in baseline Nonfarm Expenditure model.

Observation counts vary slightly due to missing data.

Descriptive Figures



Note: Sample restricted to balanced households across waves.

FIGURE A1

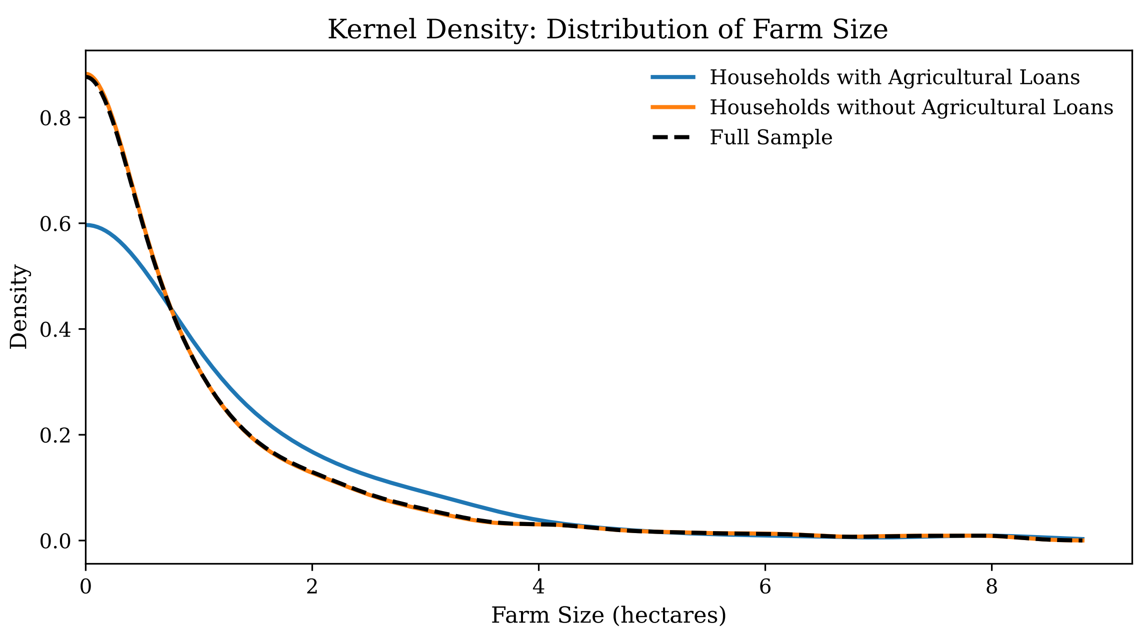


FIGURE A2

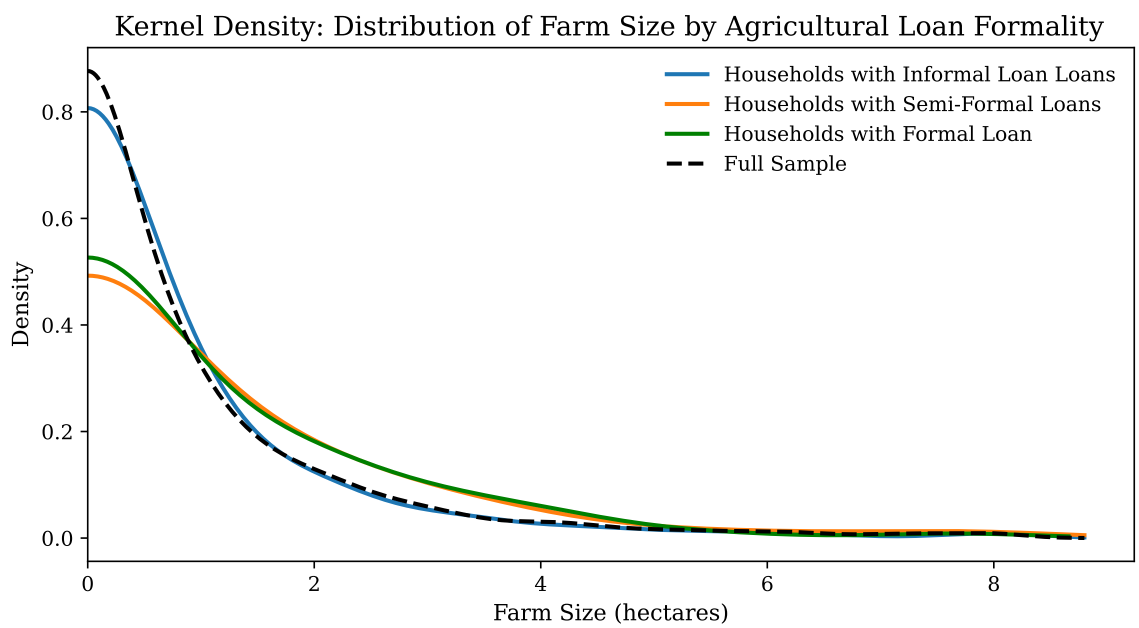

Kernel Density: Distribution of Farm Size by Agricultural Loan Formality
Households with Informal Loan Loans
Households with Semi-Formal Loans
Households with Formal Loan
Full Sample
Density
0.0
0.2
0.4
0.6
0.8
0
2
4
6
8
Farm Size (hectares)

FIGURE A3

Appendix B

The GATE estimates in Table B1 broadly align with the linear interaction reported in Table 4, though with less precision given the further dilution of the already rare treatment. In Panel B, we observe that the impact of agricultural credit on farm expenditures among the smallest landholders (Q1) is small and statistically insignificant (0.031) while the effect on the largest landholders (Q4) is large and significant at the 5% level (0.235). In contrast, Panel A shows a large effect of agricultural credit on nonfarm expenditures among the smallest landholders significant at the 10% level (0.207) with some attenuation among the largest landholders (0.112). The endpoints of the farm size distribution are therefore generally consistent with the pattern observed in our continuous specification, in which credit is highly fungible, relaxing consumption constraints among all smallholders while additionally supporting investment among larger smallholders.

The middle quartiles (Q2 and Q3), however, present results indicating a non-monotonic pattern that the continuous specification cannot capture. We attribute this partially to the loss of power: treatment effects are now identified with between 58 and 93 treated observations per quartile. Additionally, an unexpected pattern emerges between Panels A and B which is worth noting. Across quartiles, it appears treatment effects are inversely correlated, both in statistical significance and magnitude of point estimates. It should be noted that the data farm and nonfarm expenditure variables are constructed independently, and as such, this result is not a mechanical artifact of the data cleaning process. Speculatively, the inverse relationship might point directly to the fungibility of agricultural credit, reinforcing the expenditure reallocation interpretation developed in Section 4. That said, with the loss of precision in the quartile analysis, these results are ultimately inconclusive but present an opportunity for future research.

Table B1.
Group Average Treatment Effects (GATEs)
by Farm Size Quartile for Farm and Nonfarm Expenditures

	Q1	Q2	Q3	Q4
Panel A: Log Nonfarm Expenditures				
Received Agricultural Loan	0.207*	0.048	0.176*	0.112
	(0.106)	(0.086)	(0.094)	(0.100)
Treated	60	66	77	93
Observations	1,475	1,475	1,475	1,475
Panel B: Log Farm Expenditures				
Received Agricultural Loan	0.031	0.297**	−0.019	0.235**
	(0.167)	(0.149)	(0.144)	(0.109)
Treated	58	59	77	87
Observations	1,369	1,369	1,369	1,369
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes	Yes
Learners				
Loan $\sim X$	PLH-L	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R	ERH-R
K-folds	5	5	5	5

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.
GATEs estimated by interacting loan with farm size quartile indicators.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model. Diagnostics are common across quartiles within each panel.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.